

Beyond Fake Effective Sample Size: Normalizing Flow Boltzmann Sampling with Initiative Mode Discovery

Qi Feng¹, Rongjie Lai², Di Qi², and Xuda Ye²

¹Department of Mathematics, Florida State University, Tallahassee, FL 32306

²Department of Mathematics, Purdue University, West Lafayette, IN 47907

May 14, 2026

1 Introduction

The Boltzmann generator built on normalizing flows (Noé et al., 2019; Wu et al., 2020) is a central tool for sampling from complicated distributions. Let μ denote the target distribution on $\mathbb{R}^d$ and let ν denote the pushforward distribution of the flow. The conventional Boltzmann generator trains the flow by minimizing the reverse KL

$$\text{KL}(\nu\|\mu) = \int_{\mathbb{R}^d} \nu(x) \log \frac{\nu(x)}{\mu(x)} dx \quad (1)$$

with respect to the flow parameters. On multimodal targets, however, flows trained with the reverse KL are prone to mode collapse: any mode lost during training cannot be recovered, severely degrading sample quality. A standard remedy is to replace the reverse KL with the forward KL (Gabrié et al., 2022):

$$\text{KL}(\mu\|\nu) = \int_{\mathbb{R}^d} \mu(x) \log \frac{\mu(x)}{\nu(x)} dx = - \int_{\mathbb{R}^d} \mu(x) \log \nu(x) dx + \text{const}, \quad (2)$$

which is more on multimodal targets. The forward KL is the expectation of the *log-ratio* $\log(\mu/\nu)$ under μ , and therefore requires samples from the target. Such samples can be approximated by annealed importance sampling (AIS) (Midgley et al., 2022), which transports the pushforward ν toward μ through a sequence of reweighting and resampling steps. The flow AIS bootstrap (FAB) (Midgley et al., 2022) follows the same idea but trains against the more mode-sensitive weight μ^2/ν in place of μ , enhancing mode discovery.

These objectives can in principle mitigate mode collapse, but their effectiveness hinges on AIS samples that faithfully cover the target. If the pushforward ν misses modes of μ , the AIS chain—initialized at ν —is unlikely to reach them, and the resulting samples remain biased toward the modes ν already covers. This exposes a fatal defect of any AIS-reliant method, including the forward KL and FAB: the effective sample size (ESS) (Doucet et al., 2001), the gold-standard metric for assessing flow quality, can be *fake*. An ESS close to 100% does *not* guarantee mode coverage, since modes missing from ν contribute no mass to the ESS calculation, and their absence goes undetected.

This paper targets the fake ESS problem and develops a training scheme with *initiative mode discovery*, in which prior knowledge of the modes of μ is injected directly into the loss to guide the flow. Doing this safely is far from trivial: the forward KL and FAB are unbiased only when fed *accurate* samples from μ , so naively augmenting the AIS samples with the known mode locations is highly likely to corrupt the minimizer—the injected samples carry no correct importance weights and therefore pull the optimum away from μ .

The center of our training scheme is the cross-regularization functional (denoted X),

$$X_\omega(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \omega(x, y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (3)$$

where $\omega(x, y)$ is a finite positive measure on $\mathbb{R}^{2d}$ whose choice we defer to a later section. The motivation is direct: since our goal is to match ν with μ , the natural quantity to minimize is the log-ratio $\log(\mu/\nu)$ itself.

We penalize the *difference* of two log-ratios rather than a single absolute value because μ is known only up to an unknown normalization constant, and that constant cancels exactly in the difference. X offers three key benefits as a loss-function ingredient:

1. **Pointwise penalty.** The forward KL reshapes ν globally through Jensen’s inequality, whereas X penalizes log-ratio variations wherever they occur, making it sensitive to local features of the potential landscape that purely global divergences may miss.
2. **Tolerance to inaccurate target samples.** X vanishes whenever the log-ratio is pointwise constant, regardless of the accuracy of the samples drawn from μ . Inaccuracies in the AIS samples therefore barely shift the minimizer, and manually injected mode locations stay safe inside the loss—unlocking the initiative mode discovery promised above.
3. **Drop-in compatibility with the forward KL.** X shares the integrand $\log \nu$ with the forward KL, so when $\omega(x, y) = \mu(x)\mu(y)$ the same autograd graph and the same μ -samples are reused at negligible extra cost; moreover X has the same scale as the forward KL, so no additional tuning is required.

We consider two choices of the measure $\omega(x, y)$ in the X functional (3), each playing a distinct role. The first is $\omega(x, y) = \mu(x)\mu(y)$, giving

$$X_\mu(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu(x)\mu(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy. \quad (4)$$

As noted above, computing X_μ and its gradient incurs almost no extra cost beyond the forward KL, so X_μ can be added as a stability- and accuracy-improving regularizer at essentially zero overhead.

The second choice is $\omega(x, y) = \hat{\mu}(x)\hat{\mu}(y)$, sharing the same form as (4):

$$X_{\hat{\mu}}(\mu\|\nu) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \hat{\mu}(x)\hat{\mu}(y) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \quad (5)$$

where $\hat{\mu}$ need not be an accurate approximation to μ : by the second benefit of X, the minimizer of $X_{\hat{\mu}}$ is insensitive to the precise shape of $\hat{\mu}$. The sole role of $\hat{\mu}$ is to place mass on modes that the current pushforward ν has not yet reached, so a coarse mode-finder that errs on the side of broad coverage is exactly what we want. We construct such an $\hat{\mu}$ via the *Trinity* algorithm, executed in three steps:

1. Add large Brownian noise to the existing samples so that they cover a wider region of $\mathbb{R}^d$;
2. Apply LBFGS optimization to drive the noisy samples toward the mode centers of μ ;
3. Apply a short-time Langevin so that the samples spread across the local landscape of each mode.

The three steps are complementary: the Brownian noise enables discovery of modes not yet covered by ν ; LBFGS locates the mode centers precisely; the Langevin step diffuses the samples and makes $\hat{\mu}$ robust to small perturbations. The procedure also applies to targets whose mode set forms a low-dimensional manifold, since a finite-time optimization can still capture such a manifold.

Combining the forward KL with both regularizers yields our training loss for initiative mode discovery:

$$\begin{aligned} \mathcal{L}[\nu] &= \text{KL}(\mu\|\nu) + \lambda X_\mu(\mu\|\nu) + \hat{\lambda} X_{\hat{\mu}}(\mu\|\nu) \\ &= - \int_{\mathbb{R}^d} \mu(x) \log \nu(x) dx + \iint_{\mathbb{R}^d \times \mathbb{R}^d} \left(\lambda \mu(x)\mu(y) + \hat{\lambda} \hat{\mu}(x)\hat{\mu}(y) \right) \left| \log \frac{\mu(x)}{\nu(x)} - \log \frac{\mu(y)}{\nu(y)} \right| dx dy, \end{aligned} \quad (6)$$

where $\lambda, \hat{\lambda} \geq 0$ are tunable weights. To evaluate the trained flow we report ESS *together with* the coverage metric (Naeem et al., 2020), which measures how many clusters of $\hat{\mu}$ are reached by samples drawn through ν . The ESS alone can be fake when modes of μ are missing, so it must be paired with a mode-aware diagnostic: only when both ESS and coverage are high does the flow truly approximate μ .

2 Flow training with X functionals

In this section we derive the total training loss that drives the normalizing flow from the source distribution to the target, combining the forward KL with the X regularizers (4) and (5). Let U_0 and U_1 be the source and target potential functions on $\mathbb{R}^d$, and let $\mu_0 \propto \exp(-U_0)$ and $\mu_1 \propto \exp(-U_1)$ be the corresponding distributions. Given sufficient samples from μ_0 , we seek a differentiable flow map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $G_{\#}\mu_1 \approx \mu_0$; target samples are then drawn from $G_{\#}^{-1}\mu_0$ via importance sampling.

The remainder of this section is organized as follows. We first review the forward KL loss and describe how annealed importance sampling (AIS) supplies approximate samples from the target μ_1 . We then detail the computation of X_{μ} (4), including the autograd-reuse trick noted earlier. Next, we present the Trinity algorithm that constructs the wide-coverage distribution $\hat{\mu}$ used in $X_{\hat{\mu}}$ (5). Finally, we define the ESS and coverage metrics and use a minimal 2D example to demonstrate the power of the X-augmented loss.

2.1 Forward KL and annealed importance sampling

By the change-of-variables formula, the pullback density of μ_0 under G is

$$(G_{\#}^{-1}\mu_0)(y) = \mu_0(G(y)) |\det J_G(y)|, \quad y \in \mathbb{R}^d, \quad (7)$$

where $J_G(y) \in \mathbb{R}^{d \times d}$ is the Jacobian of G at y . The forward KL loss then reads

$$\begin{aligned} \text{KL}(\mu_1 \| G_{\#}^{-1}\mu_0) &= - \int_{\mathbb{R}^d} \mu_1(y) \log(G_{\#}^{-1}\mu_0)(y) dy + \text{const} \\ &= \int_{\mathbb{R}^d} \mu_1(y) \left(U_0(G(y)) - \log |\det J_G(y)| \right) dy + \text{const}. \end{aligned} \quad (8)$$

We deliberately parameterize the loss in terms of G (mapping target to source) rather than G^{-1} , so that no inversion is required during training.

Evaluating (2) requires samples from μ_1 , which are not directly available. The natural surrogate is to draw $y_1, \dots, y_B$ from the pushforward $G_{\#}^{-1}\mu_0$ and reweight by

$$w(y) = \frac{\mu_1(y)}{(G_{\#}^{-1}\mu_0)(y)} = \exp \left(U_0(G(y)) - U_1(y) - \log |\det J_G(y)| \right), \quad y \in \mathbb{R}^d, \quad (9)$$

followed by a resampling step and a short Langevin simulation to diffuse the duplicates. In high dimensions, however, μ_1 and $G_{\#}^{-1}\mu_0$ typically differ greatly, so the weights $w(y_i)$ collapse onto a few outliers and the resulting samples are of very low quality.

Annealed importance sampling (AIS) bridges this gap by inserting $M + 1$ intermediate distributions

$$\pi_k = (G_{\#}^{-1}\mu_0)^{1-\frac{k}{M}} (\mu_1)^{\frac{k}{M}} = (G_{\#}^{-1}\mu_0) \cdot w(y)^{\frac{k}{M}}, \quad k = 0, 1, \dots, M, \quad (10)$$

so that $\pi_0 = G_{\#}^{-1}\mu_0$ and $\pi_M = \mu_1$. Each rung of the ladder reweights by only $w(y)^{\frac{1}{M}}$, keeping the weight ratio close to 1 and avoiding the degeneracy of direct importance sampling:

Algorithm 1: Annealed importance sampling from $G_{\#}^{-1}\mu_0$ to μ_1

Input: samples $y_1, \dots, y_B$ from $G_{\#}^{-1}\mu_0$, ladder length M

Output: samples $y_1, \dots, y_B$ approximately distributed as μ_1

```

1 for  $k = 1, \dots, M$  do
2   Compute the per-step weights  $w(y_i)^{\frac{1}{M}}$  on  $y_1, \dots, y_B$ ;
3   Resample  $y'_1, \dots, y'_B$  from the weighted empirical distribution;
   // The resampled set has duplicates; Langevin diffuses them
4   Apply a short Langevin simulation to  $y'_1, \dots, y'_B$ , returning fresh samples  $y_1, \dots, y_B \sim \pi_k$ ;
5 end
```

2.2 Single-pass computation of forward KL and X_μ

Computing X_μ reuses almost all of the forward KL machinery. Substituting the explicit log-ratio under the flow map G (the subscript μ here refers to the target μ_1), the X_μ functional (4) becomes

$$X_\mu(\mu_1 \| G_\#^{-1} \mu_0) = \iint_{\mathbb{R}^d \times \mathbb{R}^d} \mu_1(y) \mu_1(y') \left| \left(U_0(G(y)) - U_1(y) - \log |\det J_G(y)| \right) - \left(U_0(G(y')) - U_1(y') - \log |\det J_G(y')| \right) \right| dy dy'. \quad (11)$$

Both the forward KL (2) and X_μ (11) can therefore be evaluated in a single forward pass that shares the same autograd graph:

Algorithm 2: Single-pass computation of forward KL and X_μ

Input: batch $\mathbf{y} = (y_1, \dots, y_B) \in \mathbb{R}^{B \times d}$ of samples from μ_1
Output: forward KL term and X_μ term
 // $\mathbf{z}$ carries the autograd graph through G
 1 Compute $\mathbf{z} = U_0(G(\mathbf{y})) - \log |\det J_G(\mathbf{y})|$;
 // $\mathbf{z}'$ inherits the graph of $\mathbf{z}$ at zero cost
 2 Sample a random permutation σ of $\{1, \dots, B\}$ and set $\mathbf{y}' = \mathbf{y}[\sigma]$, $\mathbf{z}' = \mathbf{z}[\sigma]$;
 3 Return $\text{mean}(\mathbf{z})$ as the forward KL term and $\text{mean}(|\mathbf{z} - \mathbf{z}' - U_1(\mathbf{y}) + U_1(\mathbf{y}')|)$ as the X_μ term;

Consequently, the only autograd work touching the flow parameters is the single backward pass through $\mathbf{z}$, identical to a vanilla forward KL step. Adding X_μ therefore incurs essentially no additional training cost.

2.3 Initiative mode discovery in $X_{\hat{\mu}}$ with the Trinity algorithm

Computing the $X_{\hat{\mu}}$ functional (5) requires samples from $\hat{\mu}_1$, the wide-coverage approximation of the target μ_1 . These samples are prepared once, before training: we generate a fixed pool of P samples $x_1, \dots, x_P$ distributed as $\hat{\mu}_1$, and at each training step we draw a mini-batch of size B from the pool and refresh it with a short Langevin chain before evaluating the loss. The pool is produced by the Trinity algorithm, which proceeds in three stages: *diffusion*, *optimization*, and *Langevin rejuvenation*.

Algorithm 3: Trinity: initiative mode discovery

Input: samples $x_1, \dots, x_P$ from μ_0
Output: samples $x_1, \dots, x_P$ forming $\hat{\mu}_1$
 // **Diffusion:** scatter the samples to a wider region of $\mathbb{R}^d$
 1 Choose a large $\sigma > 0$ and update $x_i \leftarrow x_i + \sigma \xi_i$ with $\xi_i \sim \mathcal{N}(0, I_d)$ i.i.d.;
 // **Optimization:** drive each sample toward a mode center of μ_1
 2 Run LBFGS on U_1 for a fixed number of steps to update $x_1, \dots, x_P$;
 // **Rejuvenation:** spread the samples around each mode
 3 Run a short Langevin simulation targeting μ_1 on $x_1, \dots, x_P$;

The output samples concentrate near the mode centers of μ_1 , and the initial Brownian diffusion ensures that even modes far from the support of the input are reachable. The $X_{\hat{\mu}}$ regularizer therefore penalizes any mode of μ_1 on which the current pushforward $G_\#^{-1} \mu_0$ places too little mass, driving the flow to discover modes that AIS alone could never reach.

A side effect of the diffusion step is that a large σ biases the output samples toward the outer modes—those far from the bulk of μ_0 . This bias is not only harmless but actually desirable for mode discovery, since the modes that the current pushforward fails to cover are precisely those lying far from μ_0 .

2.4 Evaluation metrics: ESS and coverage

The trained flow G induces the distribution $G_{\#}^{-1}\mu_0$ as an approximation to the target μ_1 . We assess its quality through two complementary metrics: the effective sample size (ESS), which measures how uniform the importance weights are on the support reached by the flow, and the coverage, which checks whether every mode of μ_1 has been reached at all.

The first metric is the effective sample size. Let $\mathbf{y} = (y_1, \dots, y_N) \in \mathbb{R}^{N \times d}$ be i.i.d. samples from $G_{\#}^{-1}\mu_0$, with importance weights $w(y_i)$ given by (9). The ESS is defined as

$$\text{ESS}(\mathbf{y}) = \frac{\left(\sum_{i=1}^N w(y_i)\right)^2}{N \sum_{i=1}^N w(y_i)^2} \in [0, 1], \quad (12)$$

where the upper bound 1 follows from the Cauchy–Schwarz inequality and is attained when all weights are equal. When G is perfectly trained the weights are constant and the ESS reaches 1. As emphasized in the introduction, however, this conclusion is only one-way: a high ESS certifies that the weights are uniform *on the support of $G_{\#}^{-1}\mu_0$* , but modes of μ_1 that the pushforward never reaches contribute nothing to the sum and so cannot lower the ESS—hence the *fake ESS* pitfall.

To detect such missing modes, we pair the ESS with the coverage metric of Naeem et al. (2020). Let $\mathbf{x} = (x_1, \dots, x_P) \in \mathbb{R}^{P \times d}$ be the samples of $\hat{\mu}_1$ produced by the Trinity algorithm (Algorithm 3). For each $1 \leq i \leq P$, the k -nearest-neighbor radius of x_i is

$$\text{NND}_k(x_i; \mathbf{x}) = \inf \left\{ r \geq 0 : \#\{1 \leq j \leq P : 0 < |x_i - x_j| < r\} \geq k \right\}, \quad (13)$$

i.e., the distance from x_i to its k -th nearest neighbor in $\mathbf{x}$. The coverage of $\mathbf{x}$ by the pushforward samples $\mathbf{y}$ is then

$$\text{Coverage}_k(\mathbf{y}; \mathbf{x}) = \frac{1}{P} \sum_{i=1}^P \mathbf{1} \left\{ \exists 1 \leq j \leq N : |y_j - x_i| \leq \text{NND}_k(x_i; \mathbf{x}) \right\}, \quad (14)$$

i.e., the fraction of points in $\mathbf{x}$ whose k -nearest-neighbor ball contains at least one pushforward sample. A coverage close to 1 certifies that every region populated by $\hat{\mu}_1$ has been reached by the flow, while a coverage substantially below 1 exposes precisely the missing modes that the ESS alone cannot see.

References

- Arnaud Doucet, Nando De Freitas, and Neil Gordon. An introduction to sequential monte carlo methods. In *Sequential Monte Carlo methods in practice*, pages 3–14. Springer, 2001.
- Marylou Gabri  , Grant M Rotskoff, and Eric Vanden-Eijnden. Adaptive monte carlo augmented with normalizing flows. *Proceedings of the National Academy of Sciences*, 119(10):e2109420119, 2022.
- Laurence Illing Midgley, Vincent Stimper, Gregor NC Simm, Bernhard Sch  lkopf, and Jos   Miguel Hern  ndez-Lobato. Flow annealed importance sampling bootstrap. *arXiv preprint arXiv:2208.01893*, 2022.
- Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaegun Yoo. Reliable fidelity and diversity metrics for generative models. In *International conference on machine learning*, pages 7176–7185. PMLR, 2020.
- Frank No  , Simon Olsson, Jonas K  hler, and Hao Wu. Boltzmann generators: Sampling equilibrium states of many-body systems with deep learning. *Science*, 365(6457):eaaw1147, 2019.
- Hao Wu, Jonas K  hler, and Frank No  . Stochastic normalizing flows. *Advances in neural information processing systems*, 33:5933–5944, 2020.